

Tuning of Convolutional Nowcasting Precipitation Models and Evaluation by Rainfall Event Type

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ABSTRACT: Precipitation nowcasting, the forecasting of short-term localized precipitation, is a crucial component of weather forecasting, particularly given the increasing frequency of extreme weather events due to climate change. Building on the critical role of precipitation nowcasting, this study focuses on enhancing deep learning models by optimizing hyperparameters on well-established models such as convolutional long short-term memory (ConvLSTM) and small attention U-Net (SmaAt-UNet) rather than proposing new architectures. Our dataset consists of 10 years of precipitation maps at $1 \times 1 \text{ km}^2$, 5-min resolution from Météo-France's Mosaic product covering the Paris region, providing a robust testbed for training and evaluating models. To better characterize the performance of models over different types of rain events, clustering was carried out to partition the events into four different groups ranging from light rain to intense rain. Four main changes were explored to improve the training process. We show that increasing the temporal context up to 40 min significantly improves predictive accuracy for high-intensity rainfall events. Second, the introduction of wider spatial observation windows mitigates edge effects during training, enhancing spatial consistency and intensity estimation. Third, extending the output horizon to 90 min to predict precipitation at 60 min offered substantial gains for recurrent models like ConvLSTM, highlighting their capacity to better capture long-term dynamics. Last, the influence of convolution kernel size was studied to maximize the performance of the models. These changes collectively lead to significant forecast improvements especially for extreme precipitation events with metric values increasing compared to baseline configurations. We also show that the proposed model performs better when evaluated at coarser scales, relevant for hydrological applications.

SIGNIFICANCE STATEMENT: This study examines the impact of some hyperparameters of nowcasting models on forecast performance. Through extensive experiments, we demonstrate the importance of these hyperparameters and show that forecasts can be improved in certain weather situations.

KEYWORDS: Precipitation; Nowcasting; Model comparison; Neural networks; Optimization; Deep learning

1. Introduction

Precipitation nowcasting, which is the prediction of precipitation in localized areas over the next 0–2 h, is a key task in weather forecasting. It requires not only good accuracy in estimating the amount, timing, and location of precipitation but also good estimate of small-scale spatial and temporal variations. The emergence of the concept of nowcasting is highlighted by Browning and Collier (1989), when they emphasized the importance of weather radars for forecasting precipitation, thanks to their ability to restore the spatial structure of rain with good temporal and spatial resolutions. In a recent review, Bojinski et al. (2023) refer to nowcasting as the “last line of defense” that weather services have to protect weather-sensitive assets from the worst impacts of severe weather, as a complement to longer-term forecasting. Moreover, weather patterns are altered by climate change, and as the occurrence of extreme weather events

increases, providing actionable predictions becomes more important (Marvel et al. 2023).

In recent years, approaches using optical flow (OF) models applied to quantitative precipitation estimate (QPE) deduced from radar data have been widely used. These models attempt to infer how objects will move through a sequence of images but can hardly represent dynamics such as storm onset or decay (Germann and Zawadzki 2002). Short-Term Ensemble Prediction System (STEPS) introduced by Bowler et al. (2006) is considered the baseline for nowcasting works. STEPS in its simplest version is based on the principle of Lagrangian persistence, which assumes that precipitation patterns will maintain their current state of motion and intensity over a short period of time. This persistence assumption provides a simple baseline for forecasting near-future conditions by extrapolating recent observations. However, STEPS goes beyond mere Lagrangian persistence by incorporating multiple optical flow techniques across different spatial scales to capture the evolution of precipitation fields. By combining these diverse techniques, STEPS can improve its predictive capabilities and account for various types of precipitation dynamics, providing more reliable short-term forecasts compared to basic optical flow methods.

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With the large amount of radar data provided by the operational weather radar networks, many deep learning-based methods have been developed for precipitation nowcasting in recent years. Indeed, although these models require processing large amounts of data for training, they offer low-latency inferences in operational mode, which is particularly interesting for nowcasting (NWC) where timeliness is key. As [An et al. \(2025\)](#) point out, with various perspectives, the contributions in the field of nowcasting can be synthesized in three points: optimization of models (with new architectures), the innovation in loss functions (among which some are tailored for imbalanced and sparse data), and also the enhancements in preprocessing techniques. For instance, recent works have explored weighted and pooling loss functions to better address high-intensity precipitation events and spatial patterns ([Bulo et al. 2017](#); [Ko et al. 2022](#); [Xu et al. 2024](#); [Cambier Van Nooten et al. 2023](#)). Furthermore, strategies such as sensor fusion and multivariate input models have expanded the predictive scope by integrating radar and satellite data to overcome individual limitations ([Sønderby et al. 2020](#); [Fernandez and Mehrkanoun 2021](#)).

In this paper, we focus specifically on improving the learning process of deep learning models for precipitation nowcasting. Unlike studies that look into architectural innovations, our approach emphasizes exploring the impact of hyperparameters. Given a specific problem, it seems important to fine-tune the architecture to better retrieve the dependencies in the data. Hence, we analyze how changes in the temporal and spatial contexts improve the performances for two classical models: convolutional long short-term memory (ConvLSTM) ([Shi et al. 2015](#)) and small attention U-Net (SmaAt-UNet) ([Trebing et al. 2021](#)). These results are also compared to well-known models, namely, Python STEPS (PySTEPS), the Python adaptation of the optical flow model STEPS ([Pulkkinen et al. 2019](#)), and a deep generative model of radar (DGMR) ([Ravuri et al. 2021](#)) based on another family of artificial intelligence (AI) models using generative architecture (generative adversarial networks).

Comparing models from the literature is not always an easy task due to the different datasets and the variety of preprocessing used. For example, the authors of SmaAt-UNet report that the critical success index (CSI) score can vary significantly depending on the type of preprocessing. We show that depending on the content of the databases (stratiform events vs convective events, among others), the obtained performances can be very different for a given model (convective rains being more difficult to predict than stratiform rains). To account for this, we partitioned our databases into different groups of meteorological situations allowing us to evaluate independently the performances of the models on each of these groups.

[Section 3](#) presents the dataset used, while [section 4](#) is devoted to the description of the models and evaluation metrics used. Finally, [section 5](#) focuses on comparing the two models in their original configurations and tuning specific hyperparameters to evaluate their impact on model performances.

2. Deep learning models in precipitation nowcasting

a. Recurrent models

Recurrent models are designed to capture temporal dependencies in sequential data, making them particularly well suited for nowcasting tasks where understanding the evolution of precipitation patterns over time is crucial. [Shi et al. \(2015\)](#) formulated precipitation nowcasting from radar observations as a time series of forecasting tasks and were the first to introduce a deep learning model called ConvLSTM. This model combines the strengths of convolutional neural networks for extracting spatial features ([Lecun et al. 1998](#)) and long short-term memory (LSTM) networks for modeling temporal dynamics ([Hochreiter and Schmidhuber 1997](#)) making it particularly well suited for capturing the spatiotemporal evolution of precipitation patterns. This model outperformed the model Real-Time Optical Flow by Variational Methods for Echoes of Radar (ROVER) in the Hong Kong area presented by [Woo and Wong \(2017\)](#). Despite this pioneering work with ConvLSTM, some limitations were pointed out. The dataset used for ConvLSTM contains only 97 rainy days, which seems like a small sample to capture precipitation events' variability. Moreover, the evaluation focuses solely on a 0.5 mm h^{-1} threshold that simply distinguishes rainy and nonrainy situations. However, operational systems must pay particular attention to intense, potentially damaging rainfall events, for which this threshold alone is insufficient. In a follow-up study, [Shi et al. \(2017\)](#) introduced a novel deep learning architecture, namely, the trajectory-gated recurrent unit (TrajGRU) based on convolutional-gated recurrent unit (ConvGRU) ([Cho et al. 2014](#); [Ballas et al. 2016](#)) in which the recurrent connections are dynamically determined.

b. U-Net-based encoder-decoder

While recurrent models excel in capturing temporal dependencies in sequential data, consideration of relationships across spatial scales has led to models based on specific architectures such as U-Nets. The U-Net was initially developed for biomedical image segmentation tasks ([Ronneberger et al. 2015](#)), but its unique encoder-decoder structure with skip connections makes it well suited for precipitation forecasting problems. By integrating a contracting path to capture large-scale precipitation patterns with an expansive path for precise localization, the U-Net architecture can efficiently model rainfall dynamics across multiple scales. Several studies have successfully adapted the U-Net for radar-based precipitation nowcasting, demonstrating its effectiveness compared to previous approaches. [Agrawal et al. \(2019\)](#) proposed the Radar Echo Nowcaster using an encoder-decoder architecture inspired by U-Net. Their model outperformed benchmarks, better preserving the structure and evolution of precipitation fields. Similarly, [Trebing et al. \(2021\)](#) used the SmaAt-UNet, a U-Net architecture coupled with attention mechanisms. This encoder-decoder structure allows the model to compress input information while preserving the most relevant features. Coupled with attention mechanisms, the model can focus its attention on the most important information in the data.

However, a well-known limitation of encoder–decoder networks is their tendency to produce spatially blurred forecasts, an effect that generally becomes more pronounced with longer lead times. This blurring arises from the tendency to provide an answer that evolves toward the average during training allowing to minimize the cost function in the face of increasing uncertainty. This limitation motivates the use of generative and adversarial strategies that aim to recover sharper, more physically realistic rainfall structures.

c. Generative approaches

With the advent of generative models, new approaches have emerged in precipitation forecasting. Models based on the architecture of generative adversarial networks (GANs), introduced by Goodfellow et al. (2014), have been proposed. GANs consist of two competing neural networks: A generator learns to produce realistic samples, while a discriminator attempts to distinguish these samples from real data. Applied to precipitation forecasting, the generator produces realistic future rainfall fields from previous radar observations. Thus, based on this type of architecture, Ravuri et al. (2021) trained the DGMR model on a dataset of the British Met Office radar network to obtain realistic outputs with high sharpness. This type of model effectively allows to obtain visually consistent outputs that faithfully reproduce the complex characteristics of rainfall systems. However, training GANs is computationally expensive and can be unstable.

While GANs introduced a new level of sharpness and realism to precipitation forecasting, their instability during training inspired the adoption of more robust generative frameworks, such as diffusion models, which are currently the subject of numerous studies in rain nowcasting. Diffusion models are a recent class of generative models that operate according to a reversed diffusion process. During training, the model learns to progressively remove Gaussian noise added to a starting image. Diffusion models are distinguished by their ability to model the stochasticity inherent in precipitation systems, generating realistic high-resolution images. With models such as PreDiff (Gao et al. 2023) and DiffCast (Yu et al. 2024), this is a valuable asset for accurately modeling the finer details of precipitation patterns, which are often blurred in traditional methods.

As mentioned earlier, the present work addresses several issues related to precipitation nowcasting by focusing on the influence of model hyperparameters. The influence of the length of the temporal context (number of input images) has not been well studied or justified in the literature. We also show that considering larger domains mitigates the effects at the edges of the input images and exploiting longer output horizons both have an impact on the model performances. These findings, based on better consideration of input data, could provide recommendations applicable for tuning a wide range of models, particularly to better account for the nature of rainfall, including its spatiotemporal dynamics.

3. Dataset

a. Data preprocessing

The precipitation dataset used here, hereafter named rain maps, is extracted from Météo-France's Programme ARAMIS

Nouvelles Technologie en Hydrometeorologie Extension et Renouvellement (PANTHERE) composite rain product (du Châtelet 2003; Figueras et al. 2013) for the Île-de-France region (47.9°–49.6°N; 1.0°–3.8°E; 37 024 km², corresponding to an area of 176 × 208 km²), covering 11 years from 1 January 2011 to 31 December 2021. It provides a high-quality gridded precipitation product with a spatial resolution of 1 km² and a temporal resolution of 5 min. The 5-min radar-derived rainfall accumulations are converted to instantaneous rainfall rates in millimeters per hour by multiplying the value in each pixel by 12. Nine years of data were used for the training stage, leaving out 2 years, 2011 and 2019, for the overall evaluation of the various models.

As in most rainfall datasets, maps are unbalanced, with mostly zero values (no rain) and far more light rain than heavy rain. This imbalance complicates the learning of intense precipitation by machine learning models. Therefore, to improve the dataset balance, preprocessing must be performed. The following procedure is applied to identify continuous periods of precipitation (hereafter referred to as events). They are defined as consecutive rain maps where the percentage of nonzero pixels [formally, rain rate (RR) > 0.5 mm h⁻¹] in a rain map is greater than 1%. Applying this filter resulted in a dataset composed of 1360 events lasting 1 h or more. The duration constraint is set to ensure a meaningful temporal evaluation. The final database is thus made of 25 880 rain maps. Finally, among these events, 1008 were dedicated for training, 135 were dedicated for validation, and 217 from 2011 to 2019 were kept for testing.

b. Rain event clustering

The temporal evolution of precipitation can vary depending on the type of weather system. For instance, the life cycle of a convective cell is typically around 30–40 min, while that of a stratiform event can last several hours. Therefore, the quality of forecasts can be expected to vary depending on the type of event. For this reason, the 1360 events of the dataset were partitioned to analyze model performance by event type. This approach offers a relevant perspective to better identify the respective strengths and weaknesses of the models across different meteorological situations. Evaluating performance on heavy or extreme precipitation events is essential, as these events are both the least frequent and the most difficult to predict, although they are the most likely to have a major impact on the population through flooding.

From this perspective, the chosen approach is inspired by Dilmi et al. (2017) who conducted research on rain gauges, which identified useful features for describing rainfall events. These features were adapted to the 2D rain maps and six features were selected, providing a comprehensive overview of the rainfall event. They cover different aspects such as event duration, intensity, spatial coverage, total amount of rain, variability as defined hereafter, and fluctuations in rain intensity. For each rain event i , i.e., for each continuous sequence of rain maps, the six following features were computed:

- D_e^i , duration of event i , i.e., how long it rained during this event:

$$D_e^i = T_{\text{end}}^i - T_{\text{begin}}^i \text{ (s)}, \quad (1)$$

TABLE 1. Description of the four centroids and their percentage of occurrence on the whole dataset. Bold values indicate the most significant characteristics for each event type.

	D_e^i (5 min)	$R_{\max}^i$ (mm h ⁻¹)	S_e^i	C_{RR}^i (mm)	σ_{RR}^i (mm h ⁻¹)	P_{c2}^i (mm ² h ⁻²)	% of events (% test set)	No. of events of the test set
Light small-area rain	18	2.2	6	0.11	0.32	1.6	20.3 (23)	50
Light large-area rain	34	3.9	31	0.91	0.6	10.5	28.1 (29.5)	64
Medium rain	95	14.9	27	3.45	1.15	27.9	38.2 (32.3)	70
Heavy local rain	54	131	6.5	2.85	1.98	13.7	13.4 (15.2)	33

with T_{begin}^i and T_{end}^i being the start and end time of event i , respectively.

- $R_{\max}^i$, maximal rain intensity of event i , i.e., the highest precipitation rate observed during this event:

$$R_{\max}^i = \max_{1 \leq p \leq P_{\text{tot}}, 1 \leq n \leq N_i} (RR_{p,n}^i) \text{ (mm h}^{-1}\text{)}. \quad (2)$$

Since time is discrete in the time interval $[T_{\text{begin}}^i, T_{\text{end}}^i]$, there are N_i rain maps each at a time t_n with $t_n \in [T_{\text{begin}}^i, T_{\text{end}}^i]$. The P_{tot} is the number of pixels in a rain map. Hence, $RR_{p,n}^i$ is the p th pixel of the n th map of event i .

- S_e^i , mean surface area ratio of event i , i.e., the average ratio of the surface area affected by rain during the event. This could indicate the proportion of the studied area that received rain:

$$S_e^i = \frac{1}{P_{\text{tot}} N_i} \sum_{n \in \llbracket 1, N_i \rrbracket} \sum_{p \in \llbracket 1, P_{\text{tot}} \rrbracket} I_{p,n}^i, \quad (3)$$

with $I_{p,n}^i = \begin{cases} 1, & \text{if } RR_{p,n}^i > 0.1 \text{ mm h}^{-1}, \\ 0, & \text{else,} \end{cases}$

- C_{RR}^i , mean accumulated rainfall per map over event i , i.e., the average amount of rain accumulated over the maps (or areas) during the event. This gives an idea of the total amount of rain received on the area:

$$C_{RR}^i = \frac{1}{12 P_{\text{tot}} N_i} \sum_{n \in \llbracket 1, N_i \rrbracket} \sum_{p \in \llbracket 1, P_{\text{tot}} \rrbracket} RR_{p,n}^i \text{ (mm)}. \quad (4)$$

- σ_{RR}^i , mean temporal variability across pixels for an event i , i.e., the root square of the mean of the temporal variances of the rain measured pixelwise during the event. This indicates the average temporal variability of the rain within the event:

$$\sigma_{RR}^i = \sqrt{\frac{1}{P_{\text{tot}} N_i} \sum_{p \in \llbracket 1, P_{\text{tot}} \rrbracket} \sum_{n \in \llbracket 1, N_i \rrbracket} (RR_{p,n}^i - \overline{RR}_p^i)^2} \text{ (mm h}^{-1}\text{)}, \quad (5)$$

with $\overline{RR}_p^i$ being the average of the p th pixel over event i .

- P_{c2}^i , absolute rain-rate variation at order 2 over event i , i.e., the absolute variation of the rain rate at order 2 during the event. This could indicate rapid changes or fluctuations in the intensity of the rain:

$$P_{c2}^i = \frac{1}{P_{\text{tot}} N_i} \sum_{p \in \llbracket 1, P_{\text{tot}} \rrbracket} \sum_{n \in \llbracket 1, N_i \rrbracket} |RR_{p,n+1}^i - RR_{p,n}^i|^2 \text{ (mm}^2 \text{ h}^{-2}\text{)}. \quad (6)$$

Finally, all 1360 events obtained on the selected sequences were partitioned into four groups using the k -means algorithm. Applying this partitioning on the test base then allowed us to group the 217 events into four distinct meteorological classes. Table 1 presents the main features of the four classes, and Table 2 presents their seasonality.

The centroid of the first group, comprising 20% of all events, is characterized by the lowest values for all the characteristics used. It therefore includes events that are extremely weak from a spatial or temporal point of view. This group is referred as “light small-area rain” and occurs mainly in winter (43%). A second group, referred as “light large-area rain,” occurs predominantly in winter, contains 28% of the events, and is characterized on average by longer-duration, low-intensity events with a large spatial extent. A third group, “medium rain,” includes 38% of the events and presents, on average, the longest events, with higher intensities than the previous two groups, and covers a relatively large surface area, appearing primarily in spring. A fourth group, comprising 13% of the events, consists on average of the most intense rainfall, characterized by small rain cells. This group, referred as “heavy local rain,” is representative of summer storms (76%). This partitioning into distinct groups highlights the seasonality of the events, with heavy rain concentrated mainly in summer. In appendix B, the event closest to the centroid of each group is presented briefly to obtain an overview of the events that can compose each group. Finally, the distributions of events according to the different groups are quite similar for the test dataset (in parentheses in the last column of Table 1) with that of the entire database, which guarantees the good representativeness of the scores obtained. The test set consists of 217 events in total, with 50, 64, 70, and 33 events for each respective group. While this distribution allows for a consistent evaluation across classes, the relatively small number of cases in the heavy local rain group ($N = 33$) should be noted, as it implies a lower statistical significance for the scores of this specific category compared to more populated groups.

TABLE 2. Seasonality of the four meteorological groups. Bold values indicate the predominant season for each precipitation category.

	Winter (%)	Spring (%)	Summer (%)	Autumn (%)
Light small-area rain	43	23	7	27
Light large-area rain	64	11	5	20
Medium rain	16	41	22	22
Heavy local rain	3	14	76	7

4. Methodology

a. Model configurations

The general principle is to predict a sequence of 2D precipitation maps $X_{n+1}, \dots, X_{n+K}$ at times $n + 1, \dots, n + K$ in the future, where K ranges from 12 to 24, corresponding to a prediction window of 1–2 h with a 5-min time step. The models take as input a sequence of N maps $X_{n-N+1}, \dots, X_n$, representing the current and past states with N ranging from 3 to 10 (depending on the used model) corresponding to a window of 15–50 min. All deep learning models evaluated in this study (ConvLSTM, SmaAt-UNet, DGMR) are implemented in a sequence-to-sequence fashion; i.e., they produce the entire sequence of K future rainfall maps in a single forward pass.

For this study, we selected four forecasting models widely used in the literature, each based on different principles. The first model, chosen as a reference, is based on an optical flow approach from the PySTEPS library, which implements the STEPS algorithm. This model relies on the principle of Lagrangian persistence. Although simple, this approach provides a robust basis for evaluating other models. The other three models are deep learning models that are widely recognized in the literature for short-term precipitation forecasting. The second selected architecture is based on the SmaAt-UNet model (see section 2b). The third model is based on a three-layer ConvLSTM network (see section 2a). In our case, the ConvLSTM cell is defined by Eq. (7), where “ $*$ ” denotes the convolution operator, “ $\odot$ ” denotes the Hadamard product, i_t denotes the input gate, f_t denotes the forget gate, o_t denotes the output gate, C_t denotes the memory cell, h_t is the final state, and $\mathbf{W}_{xx}$ denotes a learnable weight matrix:

$$\begin{aligned} i_t &= \sigma(X_t * \mathbf{W}_{xi} + h_{t-1} * \mathbf{W}_{hi} + b_i) \\ f_t &= \sigma(X_t * \mathbf{W}_{xf} + h_{t-1} * \mathbf{W}_{hf} + b_f) \\ o_t &= \sigma(X_t * \mathbf{W}_{xo} + h_{t-1} * \mathbf{W}_{ho} + b_o) \\ C_t &= f_t \odot C_{t-1} + i_t \odot \tanh(X_t * \mathbf{W}_{xg} + h_{t-1} * \mathbf{W}_{hg} + b_g) \\ h_t &= o_t \odot \tanh(C_t). \end{aligned} \quad (7)$$

The set of equations in (7) is a slightly simplified version from the original formulation provided by Shi et al. (2015), in which the authors incorporated the “peephole connections” introduced by Gers and Schmidhuber (2000). Unlike Shi et al. (2015), who used an architecture consisting of an encoder and a forecaster, we used a simple three-layer ConvLSTM network with convolutions based on 3×3 kernels, where the output h_n of the third layer acts as the forecaster. This layer is thus a tensor of dimensions $K \times H \times W$, with K representing the number of predicted maps and H and W representing the height and width of the maps, respectively.

The last model is the DGMR model, proposed by Ravuri et al. (2021). It is important to note that this model was trained on a dataset from the Met Office radar network. Retraining or fine tuning this model on our local dataset was not feasible due to the substantial computational resources and memory required for its large-scale generative adversarial architecture. Therefore, this model is penalized compared with models

specifically trained on a dataset from the Paris region. In particular, the relative frequency and type of heavy precipitation events as well as local topography can influence the statistical distribution of precipitation fields leading to non-optimal training for the Paris region. However, the use of this model remains relevant to illustrate the model’s generalization capacity as it was trained on rain structures and behaviors that are slightly different from those of the Paris region. This allows us to assess the robustness of the model in a slightly different geographical area, an aspect that has been little emphasized in the literature on nowcasting.

Despite their limitations, the models were chosen because they represent the base principles behind the design of most of the deep learning architectures (optical flow, encoder-decoder, recurrent networks, etc.).

b. Model training

The models were trained using an NVIDIA RTX 4090 24 GB graphics processing unit (GPU). The training time for each ConvLSTM model was approximately 6 days, varying slightly depending on the configuration [i.e., the number of input maps N (3, 4, 6, 8, 10) and the number of output maps K (12, 18, 24)], while each SmaAt-UNet model required around 3 days. From the original data, we selected sequences as described in section 3 of length $N + K$ contiguous maps. The first N maps define the input sequence, while the last K maps define the target.

Each model was trained using 5–10 random initializations, and we systematically retained the best-performing ones (even if the differences remained close).

c. Objective scores

A series of experiments were conducted to test the impact of commonly used loss functions [mean-square error (MSE), MAE, RMSE, and several weighted losses on the rain intensity like balanced MSE (BMSE)] on performances. Since it did not induce significant improvements, these results are not detailed in the present paper and we finally used the MSE loss function as in Shi et al. (2017) and Agrawal et al. (2019).

Three different metrics for each predicted map are used to assess performance. The CSI allows for the position errors to be assessed, even when the rainfall rate is not well estimated. Two thresholds have been chosen: 0.5 and 10 mm h⁻¹. The first threshold evaluates the rain cell footprint, while the second assesses the positional error of heavy rain cells:

$$\text{CSI}_k = \frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k + \text{FN}_k}. \quad (8)$$

TP, FP, and FN are true positive, false positive, and false negative, respectively, and k is the map index.

The rainfall rate MSE is calculated as

$$\text{MSE}_k = \frac{1}{P_{\text{tot}}} \sum_{p=1}^{p=P_{\text{tot}}} (\text{RR}_{p,k} - \widehat{\text{RR}}_{p,k})^2. \quad (9)$$

The $\text{RR}_{p,k}$ is the ground truth for the p th pixel of the map k , where $\widehat{\text{RR}}_{p,k}$ is the corresponding estimated value.

The Pearson coefficient, which measures linear correlations by calculating the ratio between covariances and the product of standard deviations and between actual rainfall and predicted rainfall, is given by

$$r_k = \frac{\sum_{p=1}^{P_{\text{tot}}} (\text{RR}_{p,k} - \overline{\text{RR}}_k)(\widehat{\text{RR}}_{p,k} - \overline{\widehat{\text{RR}}}_k)}{\sqrt{\sum_{p=1}^{P_{\text{tot}}} (\text{RR}_{p,k} - \overline{\text{RR}}_k)^2 \sum_{p=1}^{P_{\text{tot}}} (\widehat{\text{RR}}_{p,k} - \overline{\widehat{\text{RR}}}_k)^2}}. \quad (10)$$

The $\overline{\widehat{\text{RR}}}_k$ is the mean over all the pixels of a map k , where $\overline{\text{RR}}_k$ is the corresponding mean, computed over the pixels of the estimated map.

The frequency bias (FBI) quantifies tendencies to over- or underpredict event occurrence of rain pixels value through the use of a rainfall rate threshold and is defined as

$$\text{FBI}_k = \frac{\text{TP}_k + \text{FP}_k}{\text{TP}_k + \text{FN}_k}. \quad (11)$$

Combining these different metrics together provides an assessment of the forecast performance in terms of location, intensity, and spatial structure of the predicted rainfall fields. Furthermore, assessing the performance at different spatial scales may be useful. Indeed, for specific applications like hydrology, the legacy resolution ($1 \times 1 \text{ km}^2$) is not necessary (Aerts et al. 2022), and it may be more relevant to consider watershed scales. To this end, we will also examine the performance of the models at a resolution of $5 \times 5 \text{ km}^2$.

5. Model tuning, results, and discussion for 1-h prediction

This section, organized into five parts, is devoted to the search for hyperparameter values allowing the improvement of the scores defined above. More specifically, we compare the selected models as defined in their original paper with their modified counterparts. We first examine the impact of N (number of input rain maps) on model performance and then evaluate the benefits of expanding the size of the input maps during training. These first two points will show how a better temporal and spatial context can help achieve better results. The third part investigates extending K , the length of the output sequence. It will address how the forecast temporal horizon influences predictive accuracy. The fourth part would explore the kernel size for the convolution layers. A final part explores the combined effects of these improvements, and in the last part, the performances of the models at coarser spatiotemporal scales are evaluated.

a. Impact of input sequence length

A major challenge in designing nowcasting models is selecting the number of context images provided as input to the model. However, this issue is often overlooked in the literature, with many models arbitrarily setting this parameter to 20 min without justification. Context images, which represent the recent temporal evolution of the studied phenomenon,

TABLE 3. Mean Pearson, MSE, and CSI scores at $t + 60$ min evaluated over the test dataset for the different models with varying input sequence lengths N . The best metric scores for each model are highlighted in bold.

Models	Input sequence length (N)	Pearson score	MSE score	CSI score ($>0.5 \text{ mm h}^{-1}$)
STEPS	3	0.31	2.61	0.24
	4	0.33	2.49	0.26
	6	0.32	2.50	0.27
	8	0.33	2.54	0.27
	10	0.33	2.48	0.27
SmaAt-UNet	3	0.37	2.52	0.31
	4	0.39	2.36	0.31
	6	0.40	2.22	0.33
	8	0.40	2.13	0.33
	10	0.40	2.17	0.32
ConvLSTM	3	0.44	1.99	0.33
	4	0.44	1.94	0.34
	6	0.46	1.81	0.36
	8	0.48	1.69	0.36
	10	0.46	1.69	0.36
DGMR	4	0.39	2.50	0.32

are crucial for the model's ability to capture underlying dynamics and make accurate forecasts. However, increasing the number of context images indefinitely can be counterproductive, leading to increased model complexity and computational times without guaranteed performance improvement. Conversely, a short sequence may fail to cover the characteristic time scales of evolving precipitating systems, providing insufficient information for the model to extract relevant patterns. Investigating the impact of the number of context images on the model's predictive capabilities is essential to identify the optimal configuration balancing accuracy and computational complexity. By conducting this study on several models with different operating principles, our aim is to investigate how the number of input images affects performance with respect to the underlying dynamic processes. For each of these models (except DGMR), we analyzed the influence of the length of the input sequence from 15 to 50 min (from $N = 3$ to $N = 10$). Indeed, the progressive increase in this observation window is expected to initially improve the forecast scores, potentially reaching a plateau beyond which benefits become marginal. The obtained results (Table 3) exhibit notable discrepancies between the optical flow-based STEPS model and the deep learning models. For ConvLSTM and SmaAt-UNet models, a similar behavior is observed, with a slight improvement for all the scores up to $N = 8$ (40 min of observation). For instance, the ConvLSTM model goes from an average Pearson score of 0.44 for three input images to a score of 0.48 for eight input images. In contrast, for the STEPS model, score improvements are visible only up to $N = 3$ (15 min), after which the performance reaches a plateau more rapidly than deep learning models. This behavior can be explained by the internal mechanism of STEPS, where the number of input fields is used to diagnose advection errors, representing the uncertainty and temporal variability of the motion field (Bowler et al. 2006). These diagnosed errors are subsequently incorporated into the

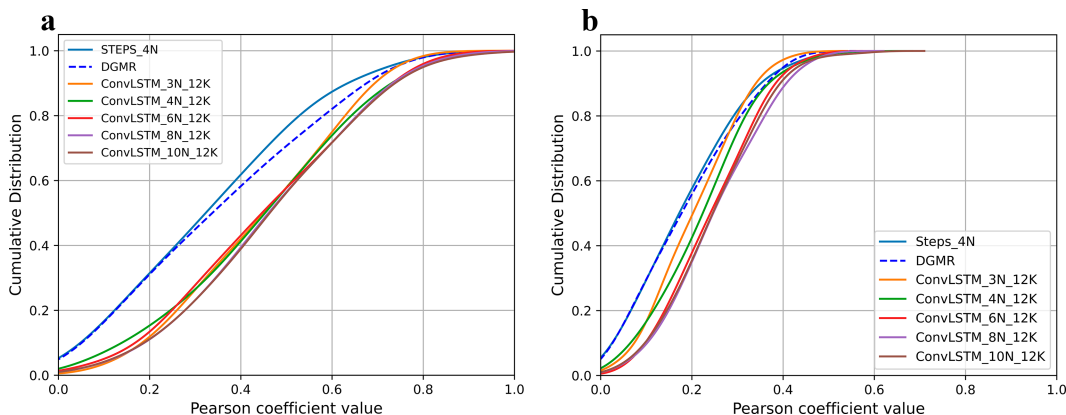


FIG. 1. CDF of Pearson's scores at $t + 60$ min for STEPS (20-min inputs) and ConvLSTM models with varying input sequence lengths N . (a) Distribution scores for the whole rainfall test dataset. (b) Distribution scores restricted to the heavy rain group.

forecast ensembles to represent the growth of advection uncertainty. Beyond this window, adding historical data fails to further reduce uncertainty in advection estimates. These results highlight a fundamental difference in how these models exploit temporal information. Finally, the best scores are obtained for $N = 8$ for both ConvLSTM and SmaAt-UNet, with a clear advantage for ConvLSTM. The standard deviation for each model remains fairly constant regardless of the number of inputs, except for the standard deviation of the MSE, which decreases for an increase in the number of inputs. Moreover, in all cases, the standard deviation remains relatively high indicating a large variability in the forecast.

To evaluate more precisely ConvLSTM models, cumulative distribution functions (CDFs) of Pearson's scores on the test dataset are shown in Figs. 1a and 1b. The CDFs in Fig. 1a were obtained on the whole test dataset, while only the heavy rain group (13.4% of the test set) was used in Fig. 1b. The ideal distribution would be a staircase-like function with zeros everywhere except where the Pearson coefficient equals 1. Performances on the whole test set show that all configurations of ConvLSTM models give more or less similar performances with a median close to 0.45 for all rain events. The left shift of STEPS and DGMR curves (median closed to 0.35) shows that all configurations of ConvLSTM models perform better. The comparison between the two figures shows that every model performance is substantially deteriorated for the heavy rain group. However, for ConvLSTM models, a shift of the curves to the right can be observed as the number of input maps increases, indicating improved forecasts. Thus, the median for the heavy rain group increases from 0.2 with a 15-min context (three input maps) to 0.26 for a 40-min context (eight input maps). However, one can notice that there is no significant difference between 8 and 10 inputs. The same behavior is displayed by the medium rain group (not shown). This behavior could be attributed to the fact that convective precipitating systems often exhibit characteristic time scales on the order of 20–30 min (NOAA2009). A 40-min context therefore seems more appropriate for capturing spatiotemporal evolution, particularly to improve the prediction of strong events.

b. Impact of extended input maps

Due to the finite size of the maps, edge effects may occur during the training phase. For example, if a rain cell appears on rain maps at time step $t + \Delta T$, the model cannot predict it from the input sequence. This leads to average predictions based on the training set and can sometimes lead to the appearance of “ghost” rain cells as shown in Fig. 3. A simple way to minimize this problem is to improve the contextual information by augmenting the input rain map size during the training phase. Having more context, the model is able to identify the rain events entering the output rain maps later in the sequence. From a practical point of view during training, we use extended (EX) rain maps (296×328 pixels) as input instead of the originals (176×208 pixels) while we keep the original size as output. This choice was made by considering an extension of 60 pixels (i.e., 60 km) applied to each of the edges of the input maps. Thus, the loss function is estimated on the original area, which means that the backpropagation of the gradient comes only from this central area, but the model is able to detect incoming rain cells in the periphery. Table 4 compares the scores obtained by the SmaAt-UNet and ConvLSTM models with $N = 8$ input maps with and without the use of extended input maps (denoted with an additional suffix EX in the case of an extended input window) during the training step. Table 4 demonstrates a significant improvement in prediction scores for both models trained with extended input maps, with a clear advantage for ConvLSTM.

More precisely, as shown in Table 4 and the Pearson CDF in Fig. 2, the ConvLSTM and SmaAt-UNet models employing the extended input rain maps exhibit superior performances over all metrics, including for heavy rainfall scenarios. The average Pearson scores at 60 min are increased from 0.48 to 0.51 for the ConvLSTM model using the extended input maps and increased from 0.38 to 0.43 for the SmaAt-UNet model.

Figure 3 shows the $t + 15$ to $t + 60$ min forecasts of a medium-intensity rain event moving eastward. This event belongs to the medium rain group with rainfall reaching up to 15 mm h^{-1} locally without weakening or strengthening during

TABLE 4. Mean scores and standard deviations (parentheses) at $t + 60$ min for STEPS, SmaAt-UNet, and ConvLSTM using the original and the EX sizes of the input maps. The best metric scores for each model are highlighted in bold.

Models	Input maps/backpropagation	Pearson score	MSE	CSI ($>0.5 \text{ mm h}^{-1}$)
STEPS ($N = 4$)	No backpropagation	0.33 (0.18)	2.54 (1.52)	0.27 (0.10)
SmaAt-UNet ($N = 8$)	Original	0.40 (0.15)	2.13 (1.23)	0.33 (0.10)
	EX	0.42 (0.15)	1.97 (1.14)	0.34 (0.10)
ConvLSTM ($N = 8$)	Original	0.48 (0.15)	1.69 (0.98)	0.36 (0.11)
	EX	0.51 (0.16)	1.66 (0.97)	0.38 (0.11)

the selected sequence of images. The top row shows the ground truth, while the other two rows show forecasts with the ConvLSTM_8N_12K and ConvLSTM_8N_12K_EX models. These models are both ConvLSTM models with 8 input maps and 12 output maps (1-h forecast). The difference between the first model (middle row of Fig. 3) and the second (bottom row of Fig. 3) is due to the use of the extended input rain maps during training. In the middle row, a ghost rain cell appears at the edge of the output rain map at $t + 30$ min (red circle). In the following maps, the cell propagates eastward. This artifact, which we also observed in other architectures like SmaAt-UNet when using restricted domains, is a result of the model's statistical expectation of inflow. In a standard configuration, the network only sees a clipped portion of the atmosphere. One plausible explanation is that during training, it learns that specific intensity and motion patterns near the boundary are statistically associated with the entry of new rain cells into the domain. The “ghost cell” is therefore a predictive hallucination statistically likely inflow generated by the model in the absence of real observations from the surrounding area. This phenomenon no longer appears at the bottom row training on extended input rain maps. In addition, this second model better estimates the shape of rain cells and the location of maxima. More precisely, for a 1-h forecast for this rain event, we obtained a CSI of 0.45 with the ConvLSTM_8N_12K, while it increased to 0.61 with the ConvLSTM_8N_12K_EX model. The same behavior is obtained for MSE, which decreases from 1.44 to 0.78.

These results highlight the benefits of adopting a wider observation window during training to circumvent boundary effects.

c. Expanding the time horizon (use of longer output sequences)

The ability to capture the spatiotemporal dynamics of rain cells is critical for precipitation forecasting. Using an extended output horizon $\Delta T'$, larger than the initial horizon of interest ΔT (with $\Delta T < \Delta T'$), can potentially force the model to better characterize the long-term dynamics. For a recurrent model, correctly estimating the state at time $t + \Delta T'$ implies that previous states have also been correctly estimated, thus improving the estimation at $t + \Delta T$. To evaluate the potential benefits of this strategy, we performed a comparative study. For each of the SmaAt-UNet and ConvLSTM models, we trained three models with three different output horizons ($K = 12, 18$, and 24) corresponding to 60, 90, and 120 min, respectively, and we evaluated the scores for a 60-min forecast.

For the SmaAt-UNet model with no recurrent component, extending the output horizon proved counterproductive. A degradation in performance can be seen in Table 5 and Fig. 4. This degradation is more pronounced the longer the horizon. In contrast, the ConvLSTM model demonstrated significant improvements on the first-hour prediction when exploiting an extended output horizon. Indeed, the optimal score is achieved for $K = 18$ ($\Delta T = 90$ min) with a Pearson score of 0.52, while it decreases to 0.48 with $K = 12$ (60 min).

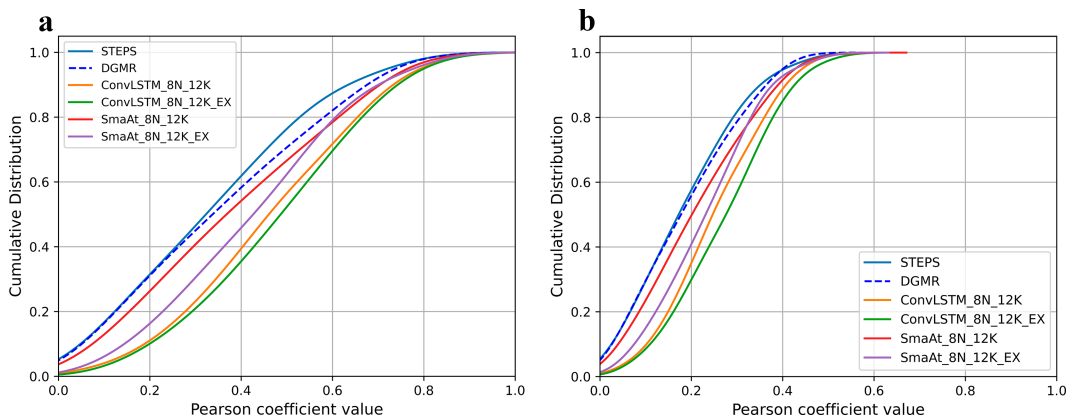


FIG. 2. As Fig. 1, but for SmaAt-UNet (40-min inputs) with standard and EX input maps and ConvLSTM under similar observation setups. (a) Distribution scores for all rainfall test events. (b) Distribution scores restricted to the heavy rain group.

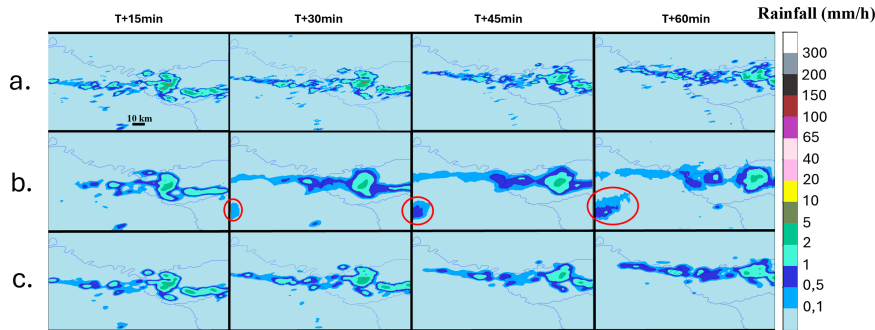


FIG. 3. Rain event occurred on 6 Jan 2011 and began around 0425 LT, containing medium intensity evolving eastward in the Paris region. (a) Ground truth, (b) prediction of ConvLSTM models trained without EX input window, and (c) ConvLSTM trained with EX input windows. The various rivers present in the region are shown as thin blue lines to give a better sense of the motion.

The recursive nature of ConvLSTM enables it to benefit from a longer output horizon, most likely because it fosters a better characterization of the underlying temporal dynamics. However, overextending this horizon beyond a certain limit (e.g., 2 h) proves counterproductive, as the accumulation of errors over very long periods counterbalances the benefits of capturing long-term dynamics. Similarly, output horizons that are too short (e.g., 1 h without additional forecast frames) do not sufficiently constrain the model, limiting its ability to learn robust temporal dynamics. Here, an intermediate horizon of 1 h 30 min seems to be an optimal compromise.

d. Size of the convolutional kernels

As explained in section 4, the ConvLSTM_8N models are built with three ConvLSTM layers, each comprising one 2D convolutional layer. Each of these convolutional layers is defined by the size of its kernel. The size of the kernel plays an important role in capturing spatial relationships, but there are no precise rules to estimate their optimal size, which depends on the type of data and the task to be solved. By default, we used small 3×3 kernels for each of the three layers, which is not necessarily optimal because rain cells can extend over several kilometers (pixels) and cannot be captured by the first layers of the network. To explore configuration possibilities, we use a grid search procedure highlighted that an optimal configuration is obtained by using kernels of sizes 7×7 , 5×5 , and 3×3 for the first, second, and third layers, respectively. These larger kernels facilitate the capture of large spatial patterns. Moreover, we found that increasing the filter size excessively

was counterproductive and it tended to smooth out rain fields too much, washing out fine details.

e. Total contributions

The model that combines all previous contributions is denoted as ConvLSTM_8N_18K_EX_3-3-3. This notation is used to emphasize the use of the 3×3 kernels for the three layers. In Fig. 5, this model (purple curve) achieves the best performance compared to previous models both on the whole test set and on the local heavy rain group, especially for small Pearson values that are less likely to occur. Thus, 25% of the Pearson values are below 0.4 compared to 36% for the other models. We also compare this model with the kernel-optimized model referred as ConvLSTM_8N_18K_EX_7-5-3 (ConvLSTM, 8 inputs, 18 outputs, extended window, kernel sizes 7×7 , 5×5 , 3×3) (brown curve). Changing the size of the first layers leads to substantial improvements in the forecast scores. This is particularly noticeable for heavy rain groups where 69% of the Pearson coefficients are below 0.4 compared to 80% for the ConvLSTM_8N_18K_EX_3-3-3 and 88% for all previous ConvLSTM models.

6. Impact of forecast horizon and spatial resolution on scores

a. 5–60-min assessment

The evolution of the scores for forecasts between 5 and 60 min is presented now. The different scores obtained for the very light and heavy rain groups as a function of time step are

TABLE 5. Mean scores and standard deviation (parentheses) at $t + 60$ min for the following model: STEPS ($N = 4$); SmaAt-UNet ($N = 8$) trained with $K = 12, 18$, and 24; and ConvLSTM with similar setups. The best metric scores for each model are highlighted in bold.

Models	Output horizon length (K)	Pearson score	MSE score	CSI ($>0.5 \text{ mm h}^{-1}$) score
SmaAt-UNet_8N	STEPS_4N	0.33 (0.18)	2.54 (1.52)	0.27 (0.10)
	12	0.40 (0.15)	2.13 (1.23)	0.33 (0.10)
	18	0.38 (0.14)	2.17 (1.26)	0.32 (0.09)
	24	0.38 (0.14)	2.22 (1.35)	0.31 (0.09)
ConvLSTM_8N	12	0.48 (0.15)	1.69 (0.98)	0.36 (0.11)
	18	0.52 (0.15)	1.61 (0.95)	0.40 (0.11)
	24	0.50 (0.14)	1.71 (0.98)	0.38 (0.11)

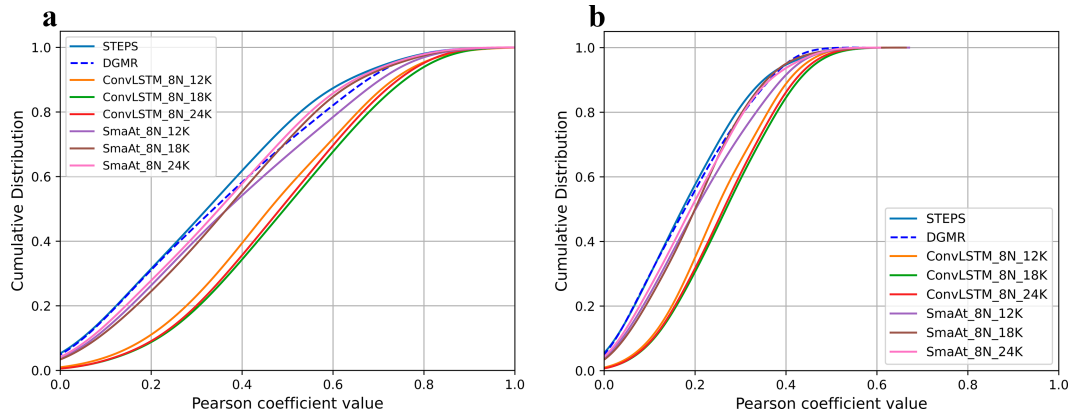


FIG. 4. As Fig. 1, but for SmaAt-UNet (40-min inputs) with 60-min ($K = 12$), 90-min ($K = 18$), and 120-min ($K = 24$) outputs and ConvLSTM with similar setups. (a) All test dataset. (b) Test dataset restricted to the heavy rain group.

presented in Fig. 6. For this final part, a frequency bias score was added to observe the blurring effect at the output of non-generative deep learning models. Only STEPS, DGMR, and the best SmaAt-UNet and ConvLSTM models are presented.

Overall, the ranking of the models remains consistent across different forecast horizons, showing a regular degradation of performance as the lead time increases. First, the analysis of the FBI reveals a common trend across all models, the understanding of which is essential for interpreting their respective performances.

As shown in Figs. 6g and 6h, all models exhibit a systematic frequency bias that evolves with the forecast horizon. For the light rain group (threshold $> 0.5 \text{ mm h}^{-1}$), the FBI increases from approximately 1.0 at 5 min to values between 1.35 and 1.5 (depending on the model) at 60 min. Conversely, for the heavy rain group (threshold $> 10 \text{ mm h}^{-1}$), the FBI decreases significantly, with the best-performing models (ConvLSTM_8N_18K_EX_7-5-3 and DGMR) reaching values around 0.71 and 0.70, respectively, while SmaAt-UNet drops to 0.66.

It is important to distinguish this bias from the visual “blurring” effect. While the spreading of rain cells due to blurring (inherent to MSE-based training) leads to the spatial expansion of

rain clusters and the subsequent underestimation of peak intensities, it is not the primary cause of the bias. This is evidenced by the DGMR model: Despite its generative constraints designed to maintain sharp textures, it follows a similar bias trend to the nongenerative models. These results suggest that the overestimation of light rain and underestimation of heavy rain are primarily driven by the underrepresentation of intense precipitation events in the training dataset and the “regression to the mean” nature of standard loss functions.

Despite these biases, considering the light rain case, for forecast horizons up to 15 min, all models provide more or less the same performances with, however, slightly better scores for SmaAt-UNet. Moreover, although DGMR was not trained on rain events specific to Paris, it shows almost the same performance as ConvLSTM models. The latter provide the best performances on all scores for forecasts greater than 15 min. For the heavy rain group, the different scores degrade more quickly for all models even if ConvLSTM models always provide the best scores. Interestingly, for CSI ($> 10 \text{ mm h}^{-1}$), the ConvLSTM_8N_18K_EX_7-5-3 model is the model performing the best for a forecast window ranging from 15 to 40 min showing its ability to better predict heavy rain cells than the

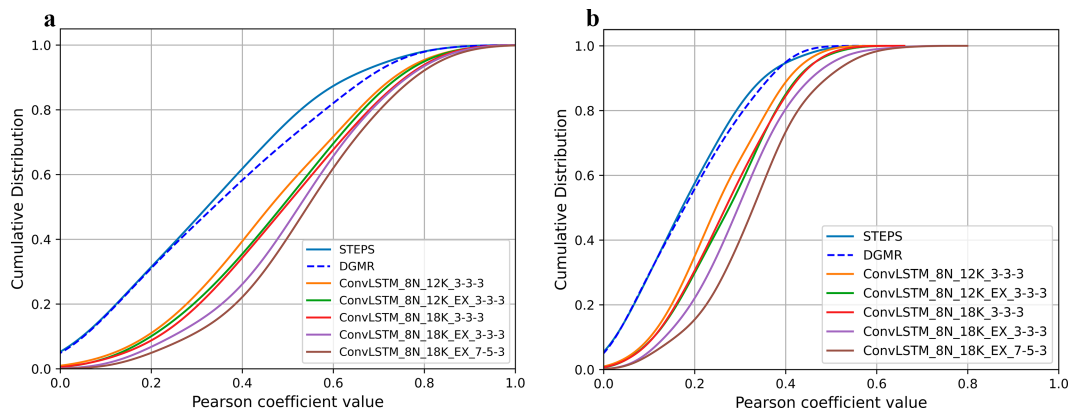


FIG. 5. As Fig. 1, but for STEPS (20 min) and ConvLSTM models under different configurations. (a) Whole test dataset. (b) Test dataset restricted to the heavy rain group dataset.

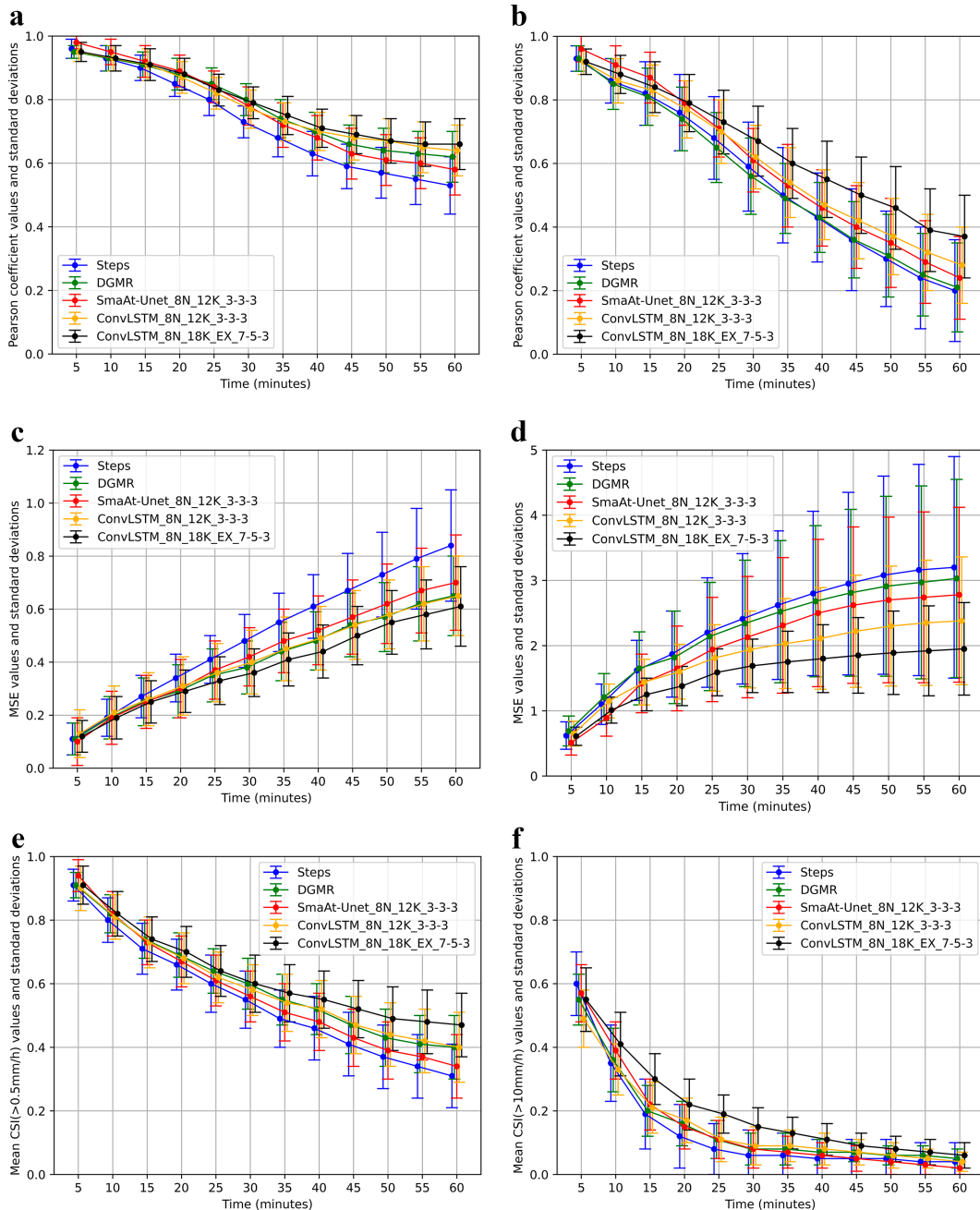


FIG. 6. Mean values (solid curves) and standard deviations (whiskers) for different metrics and models for 5–60-min forecasts. A small horizontal offset is applied to each model for each time horizon to enhance readability. (a),(c),(e),(g) The light rain group. (b),(d),(f),(h) The heavy rain group. Metrics include (a),(b) Pearson's correlation coefficient; (c),(d) MSE; (e),(f) CSI for thresholds of 0.5 (left) and 10 mm h⁻¹ (right); and (g),(h) FBI for thresholds of 0.5 (left) and 10 mm h⁻¹ (right).

other models. On the other hand, at 60 min of forecast, all models provide almost the same low CSI (>10 mm h⁻¹) score showing the difficulty of estimating the position and/or intensity of heavy rain cells beyond 1 h at the spatiotemporal resolution of 1 × 1 km² and 5 min. Finally, the scores of the other groups (light large rain and medium rain) are located between

the scores of the two groups presented in this section, as we can see, for example, on the scores of the medium rain group in [appendix A](#).

The visual impact of these behaviors is illustrated in [Figs. 7](#) and [8](#) that correspond to the forecast for STEPS, DGMR, and ConvLSTM models. The event in [Fig. 7](#), belonging to the

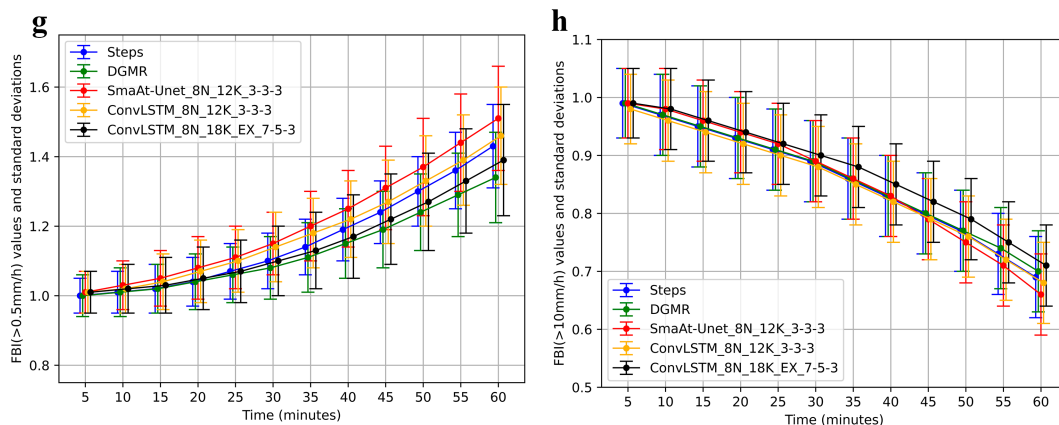


FIG. 6. (Continued)

heavy local rain group, consists of two compact rain fronts with localized convective rains (up to 70 mm h^{-1}) moving northward over the Paris region. Figure 8 shows an isolated convective cell moving eastward in the southern Paris region. It belongs to the group of “local heavy rains,” with precipitation reaching locally 120 mm h^{-1} . The STEPS and ConvLSTM_4N_12K_3-3-3 models produce outputs that lack sharpness, with blurred rain fronts. In particular, the latter merges the two rain fronts into one and struggles to maintain high-intensity precipitation in Fig. 7 and produces an increasingly spread cell for the isolated convective cell in Fig. 8. Conversely, both the DGMR and ConvLSTM_8N_18K_EX_7-5-3 models produce sharper and better-localized outputs, with CSI (0.5 mm h^{-1}) scores of 0.49 and 0.55, respectively, for 1-h forecast, compared to 0.45 and 0.41 for STEPS and the

baseline ConvLSTM_4N_12K model for Fig. 6. As explained previously, it is expected that the DGMR significantly underestimates rainfall intensity, as reflected by its higher MSE score compared to the ConvLSTM_8N_18K_EX_7-5-3 for both cases.

b. Assess at coarser scales

Although the previous results are presented for the native resolutions of 1 km^2 and 5 min, some applications use precipitation forecasts at coarser spatial and temporal scales (hydrological applications, sports events, etc.). In particular, watershed hydrological models typically use spatial resolutions on the order of a few tens of square kilometers and 1-h or more rain accumulations. In this section, we examine the performance of models when an additional postprocessing step is taken to

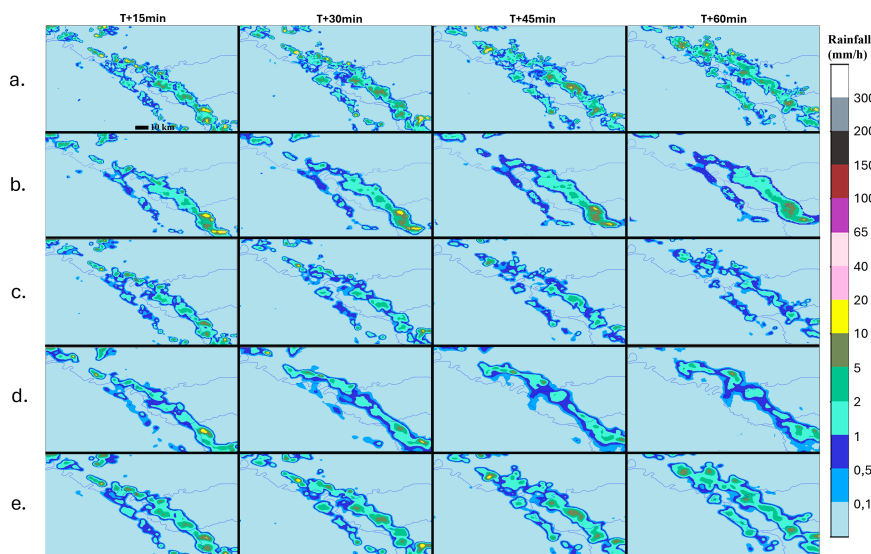


FIG. 7. Forecasts of convective cells occurred at 1600 LT 29 Mar 2011 in the Paris region. (a) Ground truth. Other rows present the forecasts of the different models. (b) PySTEPS-STEPS. (c) DGMR trained in the United Kingdom. (d) ConvLSTM_8N_12K_3-3-3. (e) ConvLSTM_8N_18K_EX_7-5-3.

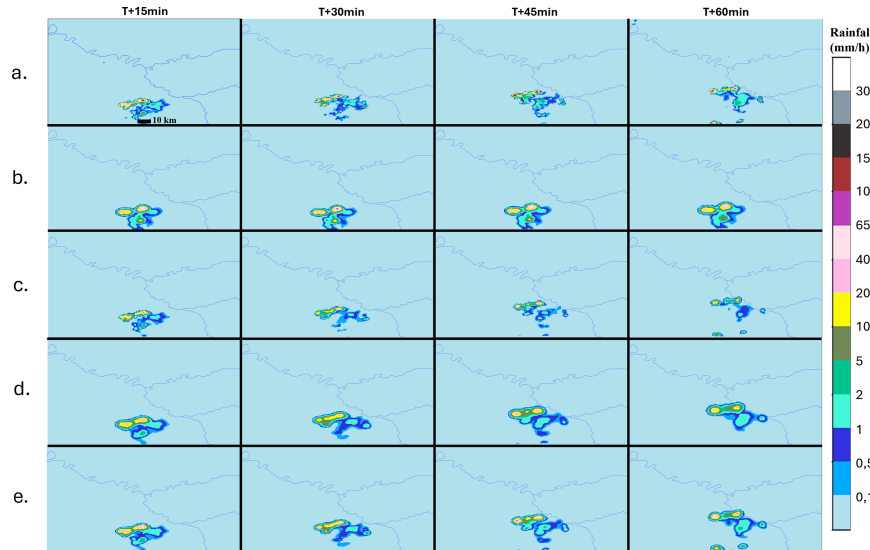


FIG. 8. Forecasts of isolated convective rain occurred at 1830 LT 21 May 2011 in the Paris region. (a) Ground truth. Other rows present the forecasts of the different models. (b) PySTEPS-STEPS. (c) DGMR trained in the United Kingdom. (d) ConvLSTM_8N_12K_3-3-3. (e) ConvLSTM_8N_18K_EX_7-5-3.

lower outputs at a coarser spatiotemporal scale. The outputs were averaged at a spatial scale of 25 km^2 (5×5 pixels) and integrated over time to obtain a 1-h rainfall accumulation. The rainfall accumulation performances provided by these models are presented in Fig. 9. Compared to Fig. 5, all curves are shifted to the right, showing a higher occurrence for high Pearson values. This was expected because the lowered spatiotemporal resolution gives less importance to the position of the rain cells and also because the rain peaks are smoothed. On the whole dataset (Fig. 9a), which contains only 13% of heavy rains, it is interesting to note that all models, except the STEPS model, have almost the same distribution, meaning that at this spatiotemporal scale, all models are more or less equivalent. For the heavy rain group (Fig. 9b), the DGMR model sees its performance decrease substantially, while the

ConvLSTM_8N_18K_EX_7-5-3 remains the best model by a slight margin.

7. Conclusions

This study focuses on tuning existing precipitation models and evaluating them on different types of rainfall events by optimizing the training configurations. Specifically, we investigate the impact of key hyperparameters including the length of the input history, the use of larger spatial rainfall maps during training, the length of the forecast output horizons, and the size of the convolutional kernels used in the 2D convolutional layers. Two deep neural networks were tuned, namely, ConvLSTM and SmaAt-UNet, while STEPS and DGMR models were used for comparison. All models use $1 \times 1 \text{ km}^2$, 5-min rain radar maps

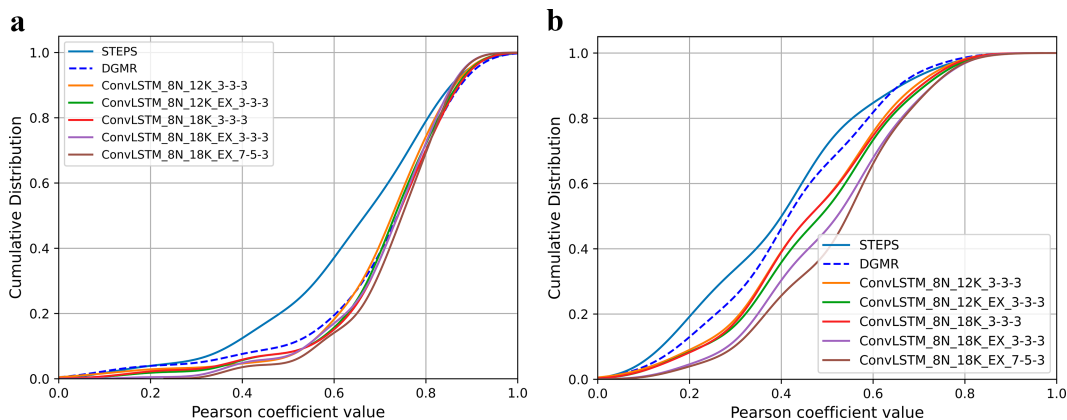


FIG. 9. As Fig. 1, but for cumulative precipitation amount over 1 h at the 25-km^2 spatial scale for the STEPS (20-min inputs) and ConvLSTM models under different configurations. (a) Whole test dataset. (b) Heavy rain group.

obtained from the Météo-France mosaic product centered on the Paris region. To evaluate the performances on different weather situations, our dataset was divided into four distinct groups. Whatever the model and its configuration, we highlight the importance of the representativeness of the used dataset. Indeed, it appears there are clear differences in performances between the four groups. This shows the difficulty of comparing models from the literature if they are not evaluated on the same dataset.

We first investigated the choice of the optimal input history length (i.e., the number of input rain maps) using a history ranging from 15 to 50 min. We show that the input history length does not play a significant role for rain events belonging to the light rain group, while a 40-min history (eight rain maps of 5 min) seems more appropriate for the heavy rain group. Moreover, the STEPS and DGMR models using a 20-min context history perform the worst regardless of the group. It should be noted, however, that the latter was trained in the United Kingdom and was not refined with our training dataset. Therefore, the comparison with DGMR is provided for informational purposes only, except to highlight the need to work in the future on the generalization of nowcasting models.

We also show that the introduction of extended input rain maps during the training not only mitigates possible edge effects, enhancing spatial consistency and intensity estimation, but also avoids the appearance in some cases of ghost rain cells. Hence, although the loss function is estimated on the original size, the model is able to account for rain cells located outside the original area. Another improvement in the case of a recurrent network is the use of a forecast window longer than the initial window of interest (in our case, 1 h 30 min for a forecast at 1 h 00 min). Extending the output horizon to 90 min allows substantial gains for a 60-min forecast, highlighting the network's ability to better capture long-term dynamics.

Deep learning models often use convolutional layers that are defined by the size of their kernels. This size is often determined empirically and commonly uses a default size of 3×3 . We showed that this default kernel size can perform well on large-scale light rain events that are rather stratiform in nature with fairly homogeneous intensity and lower dynamics. However, for heavy rain events like convective rain events, we show that the kernel size on the input layers needs to be significantly increased. In this study, we found that a size of 7×7 in the first convolutional layer provides substantial performance.

These enhancements collectively lead to marked improvements in forecast performances, particularly for extreme precipitation events. Over a 1-h period, the average Pearson score increases from 0.44 for the baseline ConvLSTM_4N_12K_3-3-3 model to 0.54 for the ConvLSTM_8N_18K_EX_7-5-3 model (a 23% improvement). Similarly, the average MSE decreases from 1.94 to 1.60 (an 18% improvement), while the CSI increases from 0.34 to 0.42 (a 24% improvement) for all rain events. For heavy rain events, the improvements are even more pronounced: The average Pearson score rises from 0.27 to 0.37 (a 37% improvement), the average MSE drops from 2.51 to 1.95 (a 22% improvement), and the CSI increases from 0.27 to 0.34 (a 26% improvement).

Importantly, these performance gains are achieved without a prohibitive increase in computational cost and memory cost.

While the optimized configuration (involving larger kernels and longer input history) slightly increases the inference time compared to the baseline deep learning model (from ~ 2.25 to ~ 2.89 s on an NVIDIA RTX 4090), it remains faster than the STEPS algorithm (~ 14.89 s on CPU) and well within the constraints of real-time operations, where a new radar map is available every 5 min. The memory usage during inference remained below 1 GB for all deep learning configurations, which fits comfortably on standard modern GPUs.

Similarly, the SmaAt-UNet model, which employs a different architectural design, has also been enhanced by optimizing the input horizon and incorporating extended input maps. While we do not detail the performance scores for this model here, these improvements demonstrate that our contributions can be generalized across various nowcasting models, regardless of their underlying architecture.

A significant finding of this study relates to the persistent challenge of frequency bias (FBI) and the blurring effect. We demonstrate that while blurring, inherent to MSE-based training, leads to the spatial expansion of rain clusters and the underestimation of peak intensities, it is not the primary cause of frequency bias. The observation that generative models like DGMR exhibit similar bias trends suggests that the overestimation of light rain and underestimation of heavy rain are primarily driven by the underrepresentation of intense events in the training data.

To address these limitations, future research should explore specialized loss functions that better emphasize extreme values, as proposed by Cambier Van Nooten et al. (2023). The application of postprocessing techniques, such as histogram matching, represents promising avenues to realign forecast distributions with observed statistics. Finally, as this study was conducted solely on the Paris region, future work should investigate the application of these improvements to various geographical contexts to confirm their robustness and adaptability to different climatic dynamics.

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Data availability statement. The 5-min rainfall accumulation data used in this study were provided by Météo-France. These data are available for academic use upon request to Météo-France. The models employed in this work are open source and can be accessed through the following GitHub repositories: PySTEPS: <https://github.com/pySTEPS/pysteps>. DGMR: https://github.com/openclimatefix/skillful_nowcasting. SmaAt-UNet (baseline model): <https://github.com/HansBambel/SmaAt-UNet>. ConvLSTM (baseline version used in this study): https://github.com/ndrplz/ConvLSTM_pytorch. The modified versions of these models, adapted for the purposes of this study, will be made publicly available on demand.

APPENDIX A

Performance Metrics for the Medium Rain Group

Figure A1 shows the performance metrics for the medium rain group, including the Pearson's coefficient (Fig. A1a), the MSE (Fig. A1b), and the CSI (Fig. A1c) for a 0.5 mm h^{-1} intensity threshold.

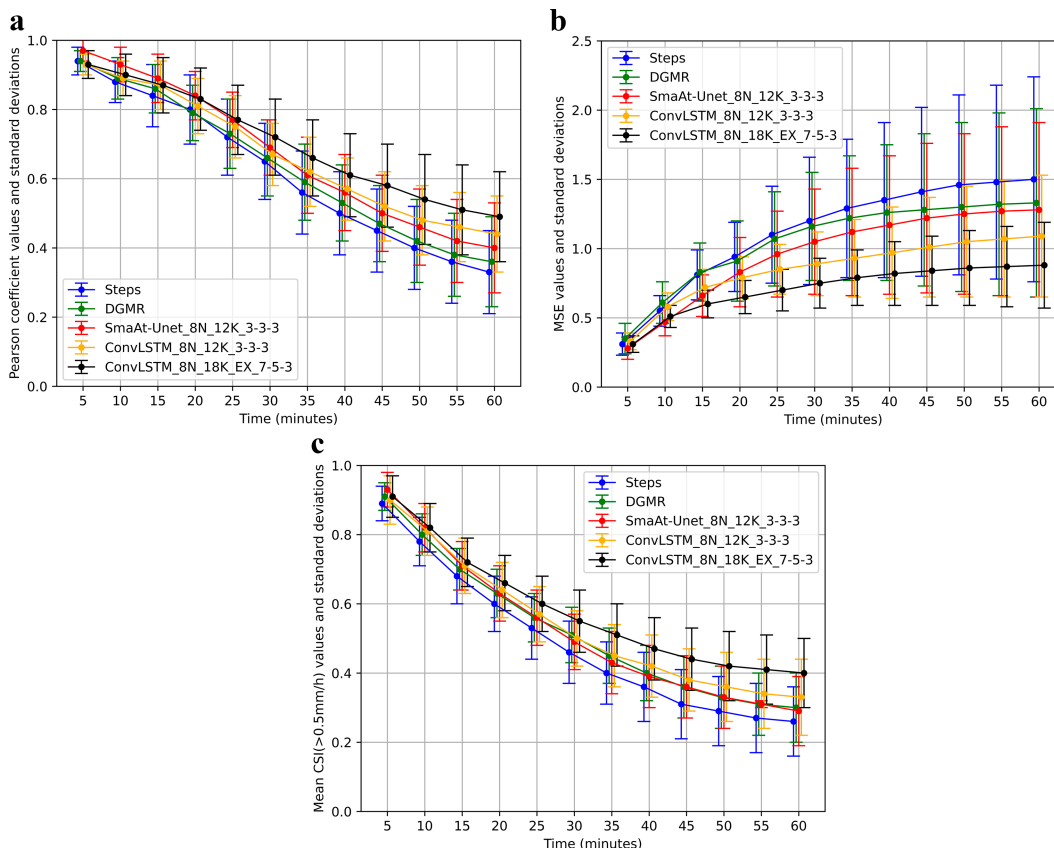


FIG. A1. (a) Pearson's coefficient scores, (b) MSE, and (c) CSI (0.5 mm h^{-1}) for the medium rain group.

APPENDIX B

Phenomenological Characteristics and Representative Examples of Precipitation Groups

The first event (Fig. B1a) began at 1600 LT 16 March 2016. This event corresponds to the spontaneous appearance of numerous small, low-intensity rain cells moving from the northeast to the southwest of the geographic area, gradually intensifying over 1 h until reaching $2.5\text{--}3 \text{ mm h}^{-1}$ in the center of the small cells. The following hour, the cells weaken and then disappear completely. This group therefore represents localized light rains, with a short duration.

The second event (Fig. B1b) is representative occurred at 0110 LT 2 February 2020 and illustrates a vast stratiform front with a large surface area and evolving slowly from west to east without much variation in rainfall intensities.

We can see the arrival of the front on the geographical area in the image. This group well represents the large winter fronts with light rainfall and a large spatial area.

As illustrated in Table 2, these two types of events are more frequent in winter. Indeed, the cold season is characterized by lower atmospheric instability: Reduced insolation and a more homogeneous air mass in temperature limit intense convection. Cold fronts and oceanic subscriptions then dominate in France, favoring the formation of extensive stratiform sheets rather than localized convective cells. These rains, although rarely intense, are not considered dangerous as extreme weather phenomena. However, their persistence in the same area can lead to lasting saturation of the ground and promote localized flooding, which is important to monitor.

The event in Fig. B1c, starting at 1730 LT 25 July 2020, is composed of several cells arriving successively to the west

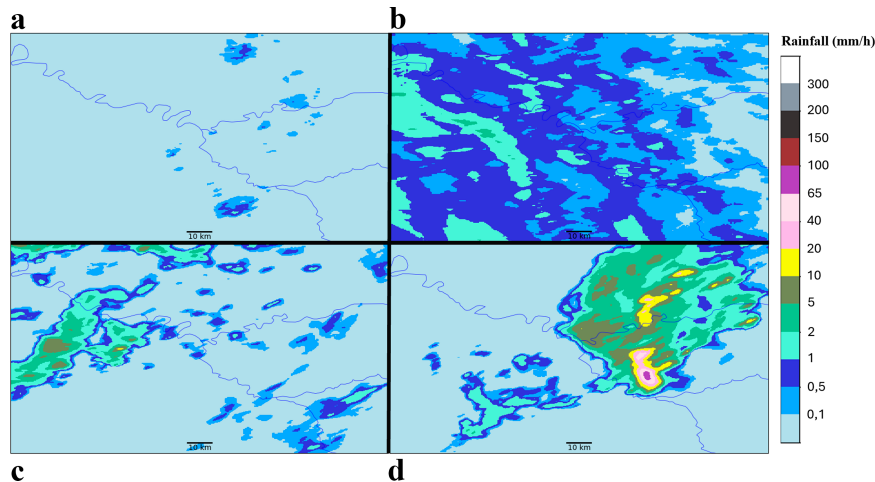


FIG. B1. Example of centroid 5-min accumulation for (a) light small-area rain, (b) light large rain, (c) medium rain, and (d) heavy local rain groups.

of the geographic area. These cells of the same nature moving eastward have rainfall intensities that can locally reach 20 mm h^{-1} , without any notable phase of weakening or intensification, with a large proportion of the cells having intensities between 2 and 5 mm h^{-1} . They leave the image to the east and mark the end of the event.

The last event (Fig. B1d) starting at 1045 LT 6 August 2019 corresponds to the passage of a precipitating system from west to east, sheltering very intense convective cores at the heart of the front, with intensities between 20 and 100 mm h^{-1} (yellow and white pixels). A peak intensity is observed at the time of the image (Fig. B1d) around 1445 LT, when up to 123 mm h^{-1} (purple pixels) is measured locally in the center.

These heavy rain events are typical of the summer season, due to high thermal instability: Maximum sunshine generates strong vertical temperature gradients, favoring deep convection. The abundant humidity in the lower layers of the atmosphere fuels the formation of very intense convective cores. In terms of impact, these intense rains can cause rapid and localized flooding.

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